Lecture 03

Introduction to Conventional Optimization Models (DP&LP) Part I


UN WATER
22 MARCH
WORLD WATER DAY
2022 Groundwater

Review on Lecture 02

❖ **Modeling Water Resources Systems**

- Models = **"Simplified Representations"** of the real-world systems
- We use models to predict (or answer "what if" questions) possible consequences (especially **quantitative**) of our actions

❖ **Types of Water Resources Systems Models**

- Optimization, Simulation, Simulation/Optimization Models

❖ **Examples of Water Resources Systems Models**

- HEC-HMS, HEC-RAS, HEC-ResSim Models
- Shared Vision Planning Model

Review on Assignment #1

1) 23 UPH 중 가장 중요하다고 생각하는 질문?
2) IWRM vs Sustainable Water Resources Management
3) Statistics regarding worldwide water resources

4) - 5) Boryeong Reservoir Model Building

- ◆ Effects on Implementation of Hedging rules: Theory and in practice?
- ◆ Key assumptions inside the model you built
- ◆ How to enhance model's practicality?

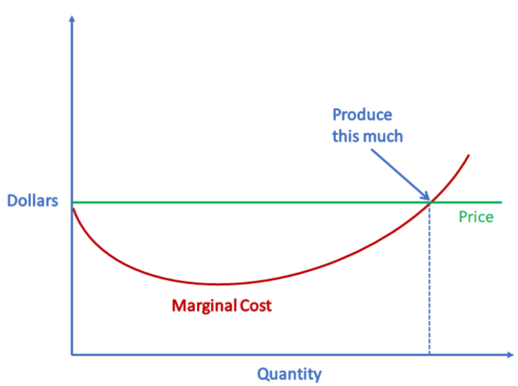
❖ DATACAMP: Classroom Plan

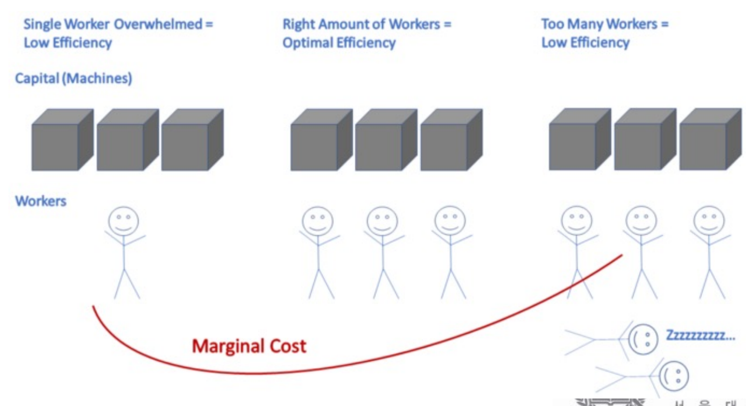
Before Lecture 03

❖ **<u>Acronyms (약어)/Keywords</u>** in Lecture 03

- ◆ **DP**: Dynamic Programming (동적 계획법)
- ◆ **LP**: Linear Programming (선형 계획법)

- ◆ **<u>Shadow Price</u>** (잠재 가격)
  - the change in the objective value of the optimal solution of an optimization problem obtained by relaxing the constraint by one unit
  - = marginal utility of relaxing the constraint
  - = marginal cost of strengthening the constraint





Today's Topic: Intro to Traditional Optimization Models

❖ Engineering Economics:
Time Streams of Economic Benefits & Costs

❖ Solving Nonlinear Optimization Models with Calculus

- ◆Solution Using Calculus
- ◆Solution Using Hill Climbing
- ◆Solution Using the Lagrange Multiplier

❖ Concepts of Dynamic Programming Models (DP)

❖ Examples of Problem Solving with DP Models

❖ Concepts of Linear Programming Models (LP)

❖ Real-world Applications of Traditional Optimization Models

Engineering Economics

❖ Key Concepts: Time Streams of Economic Benefits & Costs

◆ **Interest Rate (이자율)** = *marginal rate of return on capital*

- **Time value of money**
  (willingness to pay something to obtain money now rather than to obtain the same amount later)
- **Risk of losing capital** (not getting the full amount of a loan or investment returned at some future time)
- **Risk of reduced purchasing capability**
  (the expected inflation over time)

◆ Equivalent Present Value: converting each amount in the time series to what it is worth today (present worth)

◆ Equivalent Annual Value (EAV)

Engineering Economics

❖ 단순이자(단리)/복합이자(복리)의 차이

- ◆ 단순이자(simple interest): 이자계산기간에 원금에 대해서만 이자 발생
- ◆ <u>**복합이자(compound interest)**</u>: 매 이자계산기간의 이자를 직적기간 말에 누적된 총액에 기초하여 계산 …

❖ 기본 원리 (P/F/A)

$$F = P(1+i)^N$$

- ◆ P (현재가치)
- ◆ F (미래가치)
- ◆ A (동일자금열)

$$F = A(1+i)^{N-1} + A(1+i)^{N-2} + \cdots + A(1+i) + A$$

$$(1+i)F = A(1+i)^{N} + A(1+i)^{N-1} + \cdots + A(1+i)$$

$$F = A\left[\frac{(1+i)^N - 1}{i}\right]$$

Engineering Economics

❖ 예제 1 (Equivalent Present Value)

◆ 만약 5년 후에 확실히 3,000만원을 받거나, 지금 P원을 받을 수 있는 두 가지 대안이 있다고 가정하자. 만일 지금 P원을 받는 경우, 연 8%의 이자율을 갖는 은행계좌에 적립할 수 있다고 가정한다면, 5년 후에 3,000원만원을 받을 수 있는 약속과 지금 P원이 동일하게 되는 P 값은 얼마여야 하는가?

$F = P(1+i)^N$

◆ Sol)

$$P = F/(1+i)^N$$

$$P = \frac{3000}{(1+0.08)^5} = 2{,}042\text{만원}$$

Engineering Economics

❖ 예제 2 (Equivalent Annual Value)

- 지금부터 5년 동안 매년 말 5,000만원씩을 예금계좌에 적립하기로 하였다. 만일 이 예금계좌의 이자율이 연 6%라고 하면, 5년 말에는 얼마를 찾을 수 있겠는가?

- Sol)

$$F = A\left[\frac{(1+i)^N - 1}{i}\right]$$

$$F = 5000\left[\frac{(1+0.06)^5 - 1}{0.06}\right] = 28{,}185\text{만원}$$

Introduction to Optimization Models & Solution Procedures

❖ Mathematical Programming (Constrained Optimization)

- ◆ Calculus-based methods: ex. Lagrange multipliers
- ◆ Dynamic programming
- ◆ Quadratic programming
- ◆ Fractional programming
- ◆ Geometric programming

◆ **HIGHLY DEPENDENT** on the mathematic structure of the model (= dependent on how the system is formulated)

◆ People tend to construct models in a way that will allow them to use a particular optimization technique they think is best

Nonlinear Optimization Model: Example

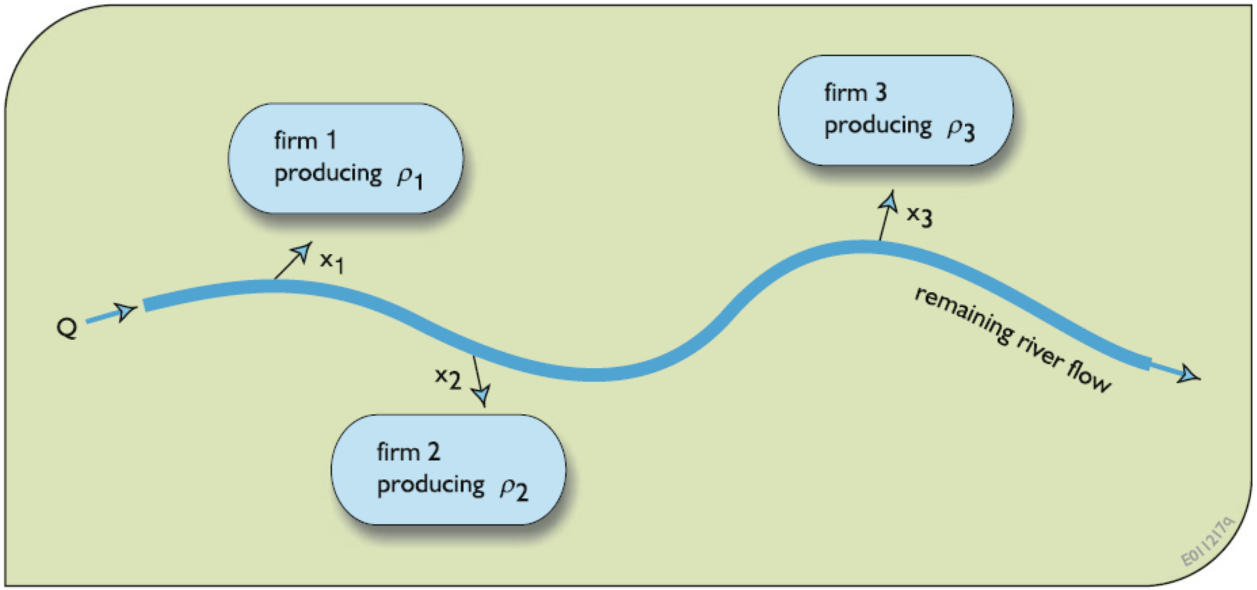


Fig. 4.1 Three water-using firms obtain water from a river. The amounts x_j allocated to each firm j will depend on the river flow Q

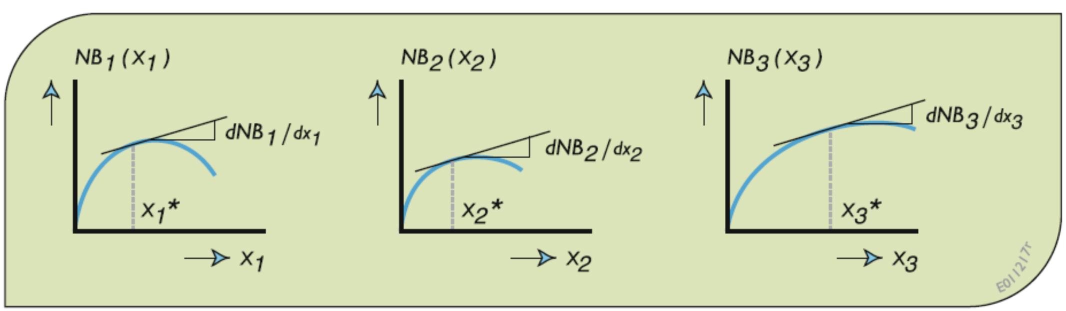


Fig. 4.2 Concave net benefit functions for three water users, j, and their slopes at allocations x_j

$$\mathrm{NB}_1(x_1) = 6x_1 - x_1^2 \tag{4.9}$$

$$\mathrm{NB}_2(x_2) = 7x_2 - 1.5x_2^2 \tag{4.10}$$

$$\mathrm{NB}_3(x_3) = 8x_3 - 0.5x_3^2 \tag{4.11}$$

Nonlinear Optimization Model: Example

❖ Approach #1: Solution Using Calculus

◆ What are Optimal values of x1, x2, and x3? Provide answers by solving the problem using calculus.

- $x_1 = 3$
- $x_2 = 2.33$
- $x_3 = 8$
- $x_1 + x_2 + x_3 = 13.33$

◆ What if $x_1 + x_2 + x_3 \leq 8$?

Nonlinear Optimization Model: Example

❖ Approach #2: Solution Using Hill Climbing

◆ What are Optimal values of x1, x2, and x3? Provide answers by solving the problem using the hill climbing method.

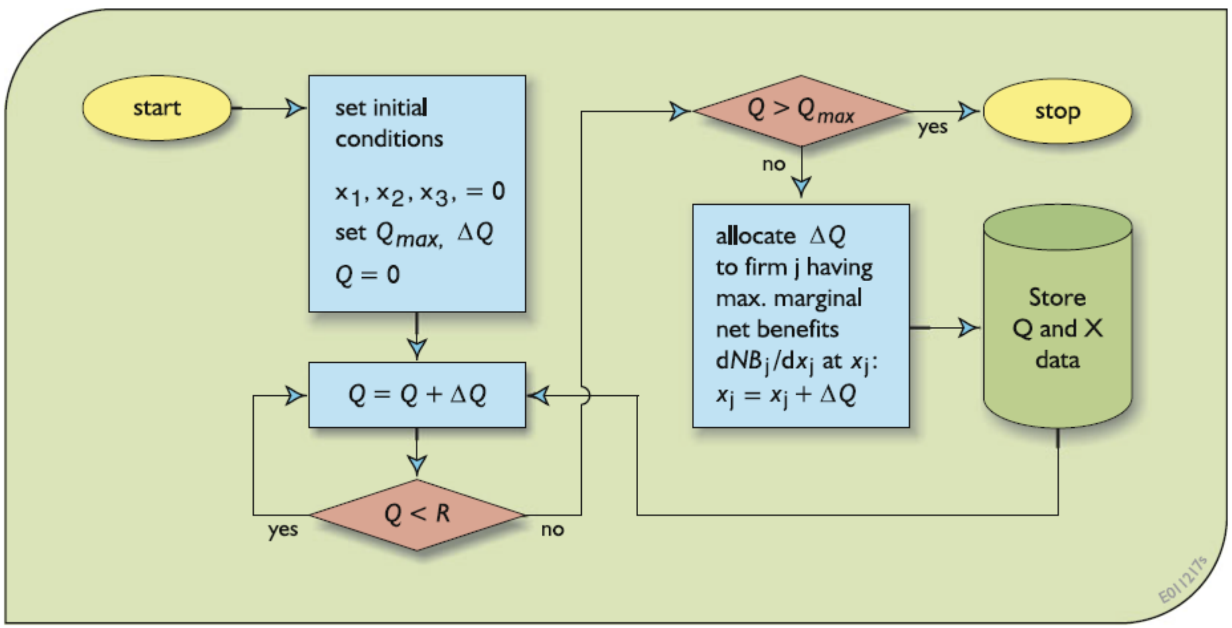


Fig. 4.3 Steepest hill-climbing approach for finding allocation of a flow Q_{max} to the three firms, while meeting minimum river flow requirements R

Nonlinear Optimization Model: Example

❖ Approach #2: Solution Using Hill Climbing

◆ What are Optimal values of x1, x2, and x3? Provide answers by solving the problem using the hill climbing method.

$Q_{max} = 8$; $Q_i = 0$; $\Delta Q = 1$; river flow $R \geq \min\{Q, 2\}$

iteration i	Q_i	allocations. R, x_j				marginal net benefits			$Q_i + \Delta Q$	new allocations				total net benefits
		R	x_1	x_2	x_3	$6-2x_1$	$7-3x_2$	$8-x_3$		R	x_1	x_2	x_3	$\Sigma_j NB_j(x_1)$
1–3	0-2	0–2	0	0	0	6	7	8	3	2	0	0	1	7.5
4	3	2	0	0	1	6	7	7	4	2	0	0	2	14.0
5	4	2	0	0	2	6	7	6	5	2	0	1	2	19.5
6	5	2	0	1	2	6	4	6	6	2	0	1	3	25.0
7	6	2	0	1	3	6	4	5	7	2	1	1	3	30.0
8	7	2	1	1	3	4	4	5	8	2	1	1	4	34.5
9	8	2	1	1	4	4	4	4	–	–	–	–	–	---

Nonlinear Optimization Model: Example

- Approach #2: Solution Using Hill Climbing
  - What are Optimal values of x1, x2, and x3? Provide answers by solving the problem using the hill climbing method.

Limitations?

-> ONLY leads to optimal allocations
only if ALL of the net benefit functions whose sum
is being maximized are **<u>concave</u>**
(=marginal net benefits decrease as the allocation increases)
Otherwise, only a local optimum is guaranteed.
(true in all calculus-based optimization procedure/algorithm)

Lagrange Multiplier λ: Example

❖ **Production Functions P(x)**

$$P_1(x_1) = 0.4(x_1)^{0.9} \quad (4.12)$$

$$P_2(x_2) = 0.5(x_2)^{0.8} \quad (4.13)$$

$$P_3(x_3) = 0.6(x_3)^{0.7} \quad (4.14)$$

❖ **Cost Functions C**

$$C_1 = 3(P_1(x_1))^{1.3} \quad (4.15)$$

$$C_2 = 5(P_2(x_2))^{1.2} \quad (4.16)$$

$$C_3 = 6(P_3(x_3))^{1.15} \quad (4.17)$$

❖ **Price of Each Units (Should be p, not ρ)**

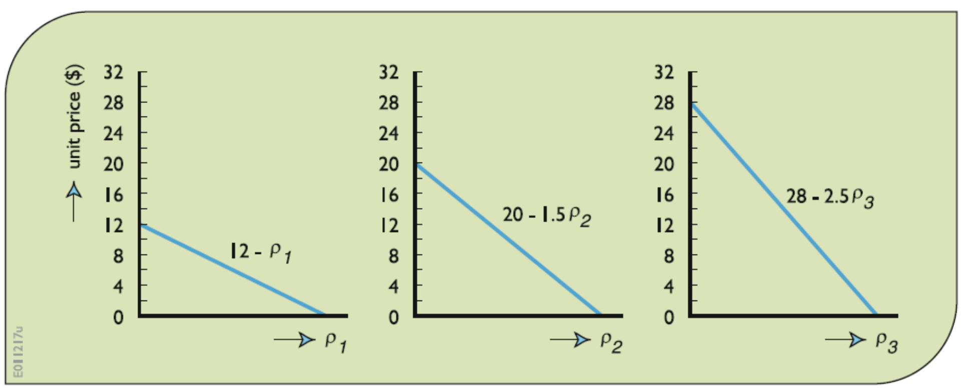


Fig. 4.5 Unit prices that will guarantee the sale of the specified amounts of products p_j produced in each of the three firms (linear functions are assumed in this example for simplicity)

Lagrange Multiplier λ: Example

❖ **Objective: Maximize Net Benefit**
(Net Benefit = Total Return – Total Cost)

$$\text{Total\_return} = (12 - p_1)p_1 + (20 - 1.5p_2)p_2 + (28 - 2.5p_3)p_3 \tag{4.20}$$

$$\text{Total\_cost} = 3(p_1)^{1.30} + 5(p_2)^{1.20} + 6(p_3)^{1.15} \tag{4.21}$$

$$\text{Total Net\_benefit} = [(12 - p_1)p_1 + (20 - 1.5p_2)p_2 + (28 - 2.5p_3)p_3] - \left[3(p_1)^{1.30} + 5(p_2)^{1.20} + 6(p_3)^{1.15}\right] \tag{4.26}$$

❖ **Subject to**

$$p_1 = 0.4(x_1)^{0.9} \tag{4.22}$$

$$p_2 = 0.5(x_2)^{0.8} \tag{4.23}$$

$$p_3 = 0.6(x_3)^{0.7} \tag{4.24}$$

$$R + x_1 + x_2 + x_3 = Q \tag{4.25}$$

Lagrange Multiplier λ: Example Sol.

❖ First, in case of an unconstrained case,

Derivatives:

$$\partial(\text{Net\_benefit})/\partial p_1 = 0 = 12 - 2p_1 - 1.3(3)p_1^{0.3} \quad (4.27)$$

$$\partial(\text{Net\_benefit})/\partial p_2 = 0 = 20 - 3p_2 - 1.2(5)p_2^{0.2} \quad (4.28)$$

$$\partial(\text{Net\_benefit})/\partial p_3 = 0 = 28 - 5p_3 - 1.15(6)p_3^{0.15} \quad (4.29)$$

The result (rounded off) is $p_1 = 3.2$, $p_2 = 4.0$, and $p_3 = 3.9$ to be sold for unit prices of 8.77, 13.96, and 18.23, respectively, for a maximum net revenue of 155.75. This would require water allocations $x_1 = 10.2$, $x_2 = 13.6$, and $x_3 = 14.5$, totaling 38.3 flow units. Any amount of water less than 38.3 will restrict the allocation to, and hence the product production at, one or more of the three firms.

❖ Solving with Lagrange Multipliers λ

1) Set up the Lagrange function $L(X, \lambda)$

$$\begin{aligned} L = {} & [(12 - p_1)p_1 + (20 - 1.5p_2)p_2 + (28 - 2.5p3)p_3] \\ & - \left[3(p_1)^{1.30} + 5(p_2)^{1.20} + 6(p_3)^{1.15}\right] \\ & - \lambda_1\left[p_1 - 0.4(x_1)^{0.9}\right] - \lambda_2\left[p_2 - 0.5(x_2)^{0.8}\right] \\ & - \lambda_3\left[p_3 - 0.6(x_3)^{0.7}\right] - \lambda_4[R + x_1 + x_2 + x_3 - Q] \end{aligned} \quad (4.30)$$

Lagrange Multiplier λ: Example Sol.

❖ Solving with Lagrange Multipliers $\boldsymbol{\lambda}$

2) Differentiate the Lagrange function L(X, λ) with respect to ten unknowns and set the resulting equations to 0:

$$\partial L/\partial p_1 = 0 = 12 - 2p_1 - 1.3(3)p_1^{0.3} - \lambda_1 \quad (4.31)$$

$$\partial L/\partial p_2 = 0 = 20 - 3p_2 - 1.2(5)p_2^{0.2} - \lambda_2 \quad (4.32)$$

$$\partial L/\partial p_3 = 0 = 28 - 5p_3 - 1.15(6)p_3^{0.15} - \lambda_3 \quad (4.33)$$

$$\partial L/\partial x_1 = 0 = \lambda_1 0.9(0.4)x_1^{-0.1} - \lambda_4 \quad (4.34)$$

$$\partial L/\partial x_2 = 0 = \lambda_2 0.8(0.5)x_2^{-0.2} - \lambda_4 \quad (4.35)$$

$$\partial L/\partial x_3 = 0 = \lambda_3 0.7(0.6)x_3^{-0.3} - \lambda_4 \quad (4.36)$$

$$\partial L/\partial \lambda_1 = 0 = p_1 - 0.4(x_1)^{0.9} \quad (4.37)$$

$$\partial L/\partial \lambda_2 = 0 = p_2 - 0.5(x_2)^{0.8} \quad (4.38)$$

$$\partial L/\partial \lambda_3 = 0 = p_3 - 0.6(x_3)^{0.7} \quad (4.39)$$

$$\partial L/\partial \lambda_4 = 0 = R + x_1 + x_2 + x_3 - Q \quad (4.40)$$

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Lagrange Multiplier λ: Example Sol.

❖ Solving with Lagrange Multipliers λ

Table 4.2 Solutions to Eqs. 4.31–4.40

water available	allocations to firms			product productions			lagrange multipliers marginal net benefits			
	1	2	3	1	2	3				
$Q-R$	x_1	x_2	x_3	p_1	p_2	p_3	λ_1	λ_2	λ_3	λ_4
10	1.2	3.7	5.1	0.46	1.44	1.88	8.0	9.2	11.0	2.8
20	4.2	7.3	8.5	1.46	2.45	2.68	4.7	5.5	6.6	1.5
30	7.5	10.7	11.7	2.46	3.34	3.37	2.0	2.3	2.9	0.6
38	10.1	13.5	14.4	3.20	4.00	3.89	0.1	0.1	0.1	0.0
38.3	10.2	13.6	14.5	3.22	4.02	3.91	0	0	0	0

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Lagrange Multiplier λ: General Form

Lagrange Multiplier λ: General Form

Lagrange Multiplier λ: General Form

Lagrange Multiplier λ: General Form

Example with Lagrange Multipliers

❖ Solve Exercise 4.13 in the Primary Textbook p.166-167.

Conclusion of Lecture 03

- **Engineering Economics:**
  Time Streams of Economic Benefits & Costs
- **Solving Nonlinear Optimization Models with Calculus**
  - **Solution Using Calculus**
  - **Solution Using Hill Climbing**
  - **Solution Using the Lagrange Multiplier**
- Concepts of Dynamic Programming Models (DP)
- Examples of Problem Solving with DP Models
- Concepts of Linear Programming Models (LP)
- Real-world Applications of Traditional Optimization Models

Future Schedule

❖ 다음 주 강의 (2022-03-29 화요일 17:00-19:50)

◆ Lec 03: Introduction to Conventional Optimization Models (DP&LP) Part II

- Engineering Economics: Time Streams of Economic Benefits & Costs
- Solving Nonlinear Optimization Models with Calculus
  - Solution Using Calculus
  - Solution Using Hill Climbing
  - Solution Using the Lagrange Multiplier
- **Concepts of Dynamic Programming Models (DP)**
- **Examples of Problem Solving with DP Models**
- **Concepts of Linear Programming Models (LP)**
- **Real-world Applications of Traditional Optimization Models**

◆Quiz #1 on Mar-29 // Datacamp #1 Due Mar-29